

1) Suppose G is finite group of even order ($\#|G| = 2m$)

Prove $\exists g \in G$, s.t. $g \neq \text{identity}$,
but $g^2 = \text{identity}$
i.e. G contains an element of order 2
 $\rightarrow$ something is its own inverse

2) Suppose $H < G$, and H consists of all the elements of G of order dividing n , for some $n \geq 2$

Prove that H is normal in G .

Let $h \in H$, $g \in G$

$$(ghg^{-1})^j = \underbrace{(ghg^{-1}) \cdots (ghg^{-1})}_{j \text{ times}} = gh^jg^{-1}$$

$$o(h) = k \quad o(ghg^{-1})$$

so $ghg^{-1} \in H$ ✓

$$\begin{aligned} gh^0g^{-1} &= e \\ gh^1g^{-1} &= g \\ gh^2g^{-1} &= g^2 \\ &\vdots \\ gh^kg^{-1} &= e \end{aligned}$$

2) Suppose $H < G$, and H is the only subgroup of G with order $n = \#|H|$.

Show $H \triangleleft G$

$h \in H, g \in G$

Show $ghg^{-1} \in H$

$\rightarrow$ Prove $ghg^{-1} \in H$, and you're there

• clearly has n elements

• clearly a group/subgroup

$$(gh_1g^{-1}gh_2g^{-1}) = gh_1h_2g^{-1} \in gh_2g^{-1}$$

• So must be H since H is only subgroup of order n

3) Suppose $H < G$ of index 2 ($|H| = \frac{1}{2}|G|$). Show $H \triangleleft G$

$g_1H = H$, g_2H ← Not subgroup
($g_1H = H \Leftrightarrow g_1 \in H$)

$$gHg^{-1}$$

$$g_1H = Hg_1$$

since neither are H

1.6 Useful for us

1.7 Group G generated by each of elements iff G is finite & prime order
 $\rightarrow$ prime order means you won't have

1.9 Non-abelian group of order 8, all subgroups normal (something I've seen before; Irish mathematicians, bridges, quaternions)

$$\{g_1, g_2, g_3, g_4, g_5, g_6, g_7, e\}$$

1.13

1.14

1.18

ALL THESE ARE QUESTIONS FOR US. I HAVE STARTED SOME BUT

WE'LL GO FROM

SCRATCH. SOME ARE FROM BOOK

AND SOME FROM DR. COX-STEIN

f adjoint $\times$ in chapter 3

$\rightarrow$ principal ideal domain (pid)

$$a, b, c \in R$$

$$(a, b, c) = \{ax + by + cz \mid x, y, z \in R\} \leq R$$

$$\begin{aligned} (7, 6) &= \{2x + by \mid x, y \in \mathbb{Z}\} \\ &= \{2x \mid x \in \mathbb{Z}\} \\ &= (2) \end{aligned}$$

$$(7, 1726) = (\gcd(7, 1726))$$

polynomials w/ coefficients of principle ideal domain

$$\mathbb{Z}[x]$$